

SAHIL AHAMED SHAIK

sahil86469@gmail.com | +61415089328 | [GITHUB](#) | [LINKEDIN](#) | [PORTFOLIO](#)

Profile

Robotics engineering student with experience in ROS 1/2, autonomous navigation, embedded systems, and robot design, currently working on ROS 2 integration and control of the Omron MoMa mobile manipulator. Skilled in developing ROS-based interfaces, sensor integration, and navigation systems for real-world robotic applications.

Skills

Programming: C/C++, Python, MATLAB

Robotics: ROS 1/2, Nav2, SLAM Toolbox, TF2, RViz, URDF, ARCL, Webots, V-REP

AI/ML: TensorFlow, CNN, ANN, Deep Learning

Design & Simulation: SolidWorks, ANSYS, Fusion 360, Simulink

Tools: Linux, Git/GitHub, VS Code, Visual Studio, Jupyter Notebook, Unity

Work Experience

Edgeforce Solutions | ROS INTERN

Dec'23 -May'24

- Extensive experience in robotics systems, including UGVs, airport tugs, and defense-related projects for the Indian Army and Indian Air Force. Specialized in integrating sensors into URDF and developing path-planning algorithms for precise navigation.
- Designed and developed various components such as UGV parts and CBM boxes, using CAD for detailed design and implementing AI & ML techniques to enhance operational capabilities.
- Proficient in ROS 1 & 2 for seamless system integration and familiar with manufacturing components for robust execution in defense applications.

Abhyaz robotics | ROBOTICS INTERN

Jun'22 - July'22

- Acquired expertise in ROS, CAD, AI & ML, and embedded systems, integral to professional growth.
- Completed projects include TurtleBot teleoperation control, basic Linux commands, AutoCAD, Tinker CAD interfaces, and UAV-related work.

Projects

ROS 2 Integration and Control of Omron MoMa Mobile Manipulator

Sep'24 – Present

- Integrating the Omron MoMa system, consisting of an LD-250 mobile robot and TM12-S arm, with ROS 2 for autonomous navigation and manipulation.
- Developed ARCL-based communication to convert robot data into ROS 2 topics such as odometry, laser scan, and point cloud.
- Configuring SLAM, Nav2, TF frames, costmaps, and localization for real-world autonomous navigation.
- Working on TMflow-based arm and gripper control for mobile manipulation tasks.
- Technologies: ROS 2, Python/C++, ARCL, Nav2, SLAM Toolbox, TF2, RViz.

Design and development of Borewell robot

Jan'24

- Developed a borewell robot emphasizing mechanical design and embedded systems to ensure efficient operation in borewell environments. Involved extensively in 3D modeling and simulations to design and validate the robot's functionality and performance.
- Applied expertise in Arduino for embedded systems development, sensor integration, real-time control, and communication protocols to enhance reliability and effectiveness.
- Technologies used: SolidWorks, ANSYS, Arduino, and various sensors.

Design and development of knee rehabilitation robot

Aug'22

- Created a knee rehabilitation robot for knee replacement and stroke patients, utilizing a less mechanical approach to ensure low energy consumption and portability
- Focused on mechanical design to reduce complexity and enhance stability, while adding control systems for accurate control and positioning during rehabilitation exercises.
- Integrated AI with Convolutional Neural Networks (CNN) to predict knee abnormalities and process patient data.
- Technologies used: Google Colab (TensorFlow), Python, SolidWorks, ANSYS

Design and development of a homogeneous swarm of cooperative robots in a virtual environment

Jan'22

- Simulated a leader-follower TurtleBot3 in the virtual environment Webots to establish intercommunication between the bots. Used Python and TurtleBots in Webots to exchange information between the nodes of bots and a centralized network.
- Technologies used: Webots, Python, MATLAB.

Education

University of New South Wales, Sydney, Australia	2024 - 2026
Master of Engineering Science (Robotics)	
Hindustan Institute of Technology and Science, Chennai, India	2020 - 2024
Bachelor of Technology in Mechatronics Engineering	

Academic and Extracurricular Achievements

- Published a paper in AIP Conference Proceedings (Volume 2946, Issue 1) and presented at RIACT 2022, RANE 2023, and AIIMS-3.0.
- Won first place in paper presentations at multiple universities – IITM, VIT, MIT (Chennai).
- Completed training: 3 days of SMC training in pneumatics, TensorFlow 2.0: Deep Learning and Artificial Intelligence, ROS for Beginners: Basics, Motion, and OpenCV, Introduction to AI, Machine Learning, and Python basics, and MATLAB and Simulink Onramp through MathWorks training services.
- Won first place in the Football interdepartmental competition in 2023.

References

Available upon request.